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Link: <https://studio.edgeimpulse.com/studio/948513/impulse/1/learning/keras/3>
Task 1

Set the 'expected outcome' for each sample to the desired outcome to automatically score the impulse.

SAMPLE NAME	EXPECTED OUT...	LENG...	ACCURACY	RESULT	
iris_normal...	virginica	1s	0%	1 uncertain	⋮
iris_normal...	virginica	1s	0%	1 versicolor	⋮
iris_normal...	virginica	1s	100%	1 virginica	⋮
iris_normal...	virginica	1s	0%	1 uncertain	⋮
iris_normal...	virginica	1s	0%	1 uncertain	⋮
iris_normal...	virginica	1s	100%	1 virginica	⋮
iris_normal...	virginica	1s	0%	1 versicolor	⋮
iris_normal...	virginica	1s	100%	1 virginica	⋮
iris_normal...	virginica	1s	100%	1 virginica	⋮
iris_normal...	virginica	1s	100%	1 virginica	⋮

Classifying data for Classifier...
Classifying data for float32 model...
✓ Job scheduled at 03 Apr 2026 00:56:01
✓ Job started at 03 Apr 2026 00:56:02
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
Classifying data for Classifier OK

Generating model testing summary...
Finished generating model testing summary

Job completed (success)

Results

Model version: ② Unoptimized (float32)

 ACCURACY
80.65%

Metrics for Classifier

METRIC	VALUE
Area under ROC Curve ②	0.99
Weighted average Precision ②	0.93
Weighted average Recall ②	0.90
Weighted average F1 score ②	0.91

Confusion matrix

	SETOSA	VERSICOLOR	VIRGINICA	UNCERTAIN
SETOSA	100%	0%	0%	0%
VERSICOLOR	0%	85.7%	0%	14.3%
VIRGINICA	0%	15.4%	84.6%	0%

Neural Network settings

Training settings

Number of training cycles ②

Use learned optimizer ② ☐

Learning rate ②

Training processor ② CPU

Advanced training settings

Neural network architecture

Neural network Complete architecture / classic ML

Input layer (4 features)

⊕ Flatten layer

Add an extra layer

Output layer (3 classes)

Training output

Model

Model version: ② Quantized (int8)

Last training performance (validation set)

 ACCURACY
100.0%

 LOSS
0.16

Confusion matrix (validation set)

	SETOSA	VERSICOLOR	VIRGINICA
SETOSA	100%	0%	0%
VERSICOLOR	0%	100%	0%
VIRGINICA	0%	0%	100%
F1 SCORE	1.00	1.00	1.00

Metrics (validation set)

METRIC	VALUE
Area under ROC Curve ②	1.00
Weighted average Precision ②	1.00
Weighted average Recall ②	1.00
Weighted average F1 score ②	1.00

Data explorer (full training set) ②

Task 2

Use learned optimizer ☐

Learning rate

Training processor

Advanced training settings

Neural network architecture

Neural network Complete architecture / classic ML

Input layer (4 features)

Flatten layer

Dense layer (20 neurons)

Dropout (rate 0.1)

Dense layer (15 neurons)

Add an extra layer

Save & train

```
Calculating tensorflow time on
Calculating float32 accuracy...
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
Calculating int8 accuracy...
Extracting TensorBoard logs...
Extracting TensorBoard logs OK
```

Model training complete

Job completed (success)

Model

Model version: Quantized (int8) ▾

Last training performance (validation set)

ACCURACY
100.0%

LOSS
0.00

Confusion matrix (validation set)

	SETOSA	VERSICOLOR	VIRGINICA
SETOSA	100%	0%	0%
VERSICOLOR	0%	100%	0%
VIRGINICA	0%	0%	100%
F1 SCORE	1.00	1.00	1.00

Metrics (validation set)

Metric	Value
Area under ROC Curve ⑦	1.00
Weighted average Precision ⑦	1.00
Weighted average Recall ⑦	1.00
Weighted average F1 score ⑦	1.00

✔ This lists all test data. You can manage this data through [Data acquisition](#).

Test data

 Classify all

Set the 'expected outcome' for each sample to the desired outcome to automatically score the impulse.

SAMPLE NAME	EXPECTED OUT...	LENG...	ACCURACY	RESULT
iris_normal...	virginica	1s	100%	1 virginica
iris_normal...	virginica	1s	100%	1 virginica
iris_normal...	virginica	1s	100%	1 virginica
iris_normal...	virginica	1s	100%	1 virginica
iris_normal...	virginica	1s	100%	1 virginica
iris_normal...	versicolor	1s	100%	1 versicolor
iris_normal...	versicolor	1s	100%	1 versicolor
iris_normal...	versicolor	1s	100%	1 versicolor
iris_normal...	versicolor	1s	100%	1 versicolor

Model testing output

```
Classifying data for Classifier...
Classifying data for float32 model...
✓Job scheduled at 03 Apr 2026 01:02:48
✓Job started at 03 Apr 2026 01:02:50
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
Classifying data for Classifier OK
```

Generating model testing summary...

Finished generating model testing summary

Job completed (success)

Results

Model version: ? Unoptimized (float32) ▾

 ACCURACY
100.00%

Metrics for Classifier

Metric	Value
Area under ROC Curve ⑦	1.00
Weighted average Precision ⑦	1.00
Weighted average Recall ⑦	1.00
Weighted average F1 score ⑦	1.00